variable potency. Potency refers to the number of possible fates that a cell can acquire. The two important properties of stem cells that differentiate them from other cells are:

i. its ability to divide and give rise to more stem cells by self-renewal

ii. its ability to give rise to specialised cells with specific functions by the process of differentiation.

Types of stem cells

Embryonic stem cells can be extracted and cultured from the early embryos. These cells are **derived from the inner cell mass of blastocyst**. These cells can be developed into any cell in the body.

Adult stem cell or **somatic stem cell** are found in the neonatal (new born) and adults. They have the ability to divide and give rise to specific cell types. Sources of adult stem cells are amniotic fluid, umbilical cord and bone marrow.

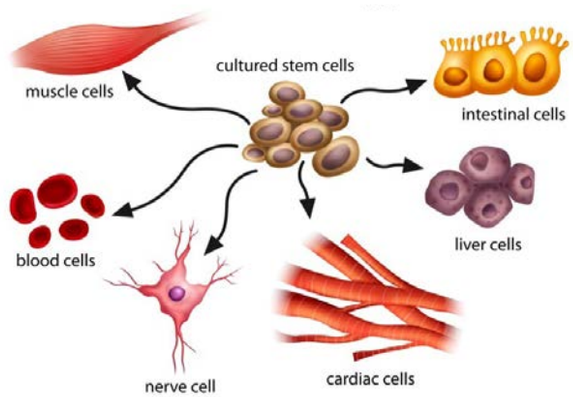


Figure 20.8 Differentiation of stem cells

Stem-cell therapy

Sometimes cells, tissues and organs in the body may be permanently damaged or lost due to genetic condition or disease or injury. In such situations stem cells are used for the treatment of diseases which is called **stem-cell therapy**. In treating **neurodegenerative disorders** like Parkinson's disease and Alzheimer's disease **neuronal stem cells** can be used to replace the damaged or lost neurons.

20.8 DNA Fingerprinting Technology

The human genome has 3 billion base pairs. Did you know that the DNA pattern of two individuals cannot be same except for identical twins. Each person's DNA sequence is unique due to the small difference in the base pairs. Therefore, if we want to compare the genetic difference among the two individuals, **DNA fingerprinting** is the easier and quicker method. This technique was developed by **Alec Jeffrey**.

The technique analyses each individual's unique DNA sequences and provides distinctive characteristics of individual which helps in identification. **Variable number of tandem repeat sequences** (VNTRs) serve as molecular markers for identification.

In human beings, 99 % of the DNA base sequences are the same and this is called as bulk genomic DNA. The remaining 1 % DNA sequence differs from one individual to another. This 1 % DNA sequence is present as small stretch of repeated sequences which is known as **satellite DNA.** The number of copies of the repeat sequence also called as **VNTRs** differs from one individual to another, and results in variation in the size of the DNA segment.

```
Person 1 ..GCCAGCTAGCTAGCTAGCTAGCTAGCTTTCAT..
            1   2   3   4   5   6

Person 2 ..GCCAGCTAGCTAGCTAGCTAGCTTTCAT..
            1   2   3   4   5

Person 3 ..GCCAGCTAGCTAGCTAGCTAGCTAGCTAGCTT..
            1   2   3   4   5   6   7
```

VNTRs illustration of three persons

As shown in the illustration, the sequence AGCT is repeated six times in first person, five times in second person and seven times in third

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